Do crowded neighbourhoods drag ratings down? Evidence from Bengaluru restaurant listings

Everything in light blue is a worked example: replace it with your own. Its numbers are made up for illustration. Grey italic lines are instructions: delete them, or hide them all with `\showhintsfalse` at the top of the file, where you also set your project title. Sections 1–5 follow items 1–5 of the report guidelines; Section 6 is the limitations of your analysis. At most 8 pages.

Group name Team Kurtosis

Dataset kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants

Code github.com/team-kurtosis/cse558-midterm (notebook and README)

Code: a link to the code written by you and a readme file, which anyone with the link can open. Also upload your notebook or script on the classroom.

#	Name	Roll no.	Institute email	Contribution
1	Aditi Rao	2022145	aditi22145@iiitd.ac.in	Dataset and size check (1)
2	Rohan Mehta	2022311	rohan22311@iiitd.ac.in	Preparation, before and after (2, 3)
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4	Karan Bose	2022260	karan22260@iiitd.ac.in	Test 1 (5)
5	Meera Nair	2022019	meera22019@iiitd.ac.in	Test 2, limitations, code and README (5, 6)

1. Dataset

Where you have collected the dataset, how many rows and columns there are, and whether #instances × #attributes is at least 3M after encoding (show the arithmetic). Name some features with their interpretation to the real world.

Source: Kaggle, "Zomato Bangalore Restaurants": scraped restaurant listings, downloaded on 28 August 2026. One row is one outlet.

What is raw about it: ratings are stored as text (`"4.1/5"`, `"NEW"`, `"-"`); the cost for two is a string with thousands separators; `dish_liked` is a comma-separated list; some chain outlets are listed twice; 15% of ratings are missing or `NEW`.

	Raw file	After encoding
Instances (rows)	51,717	48,304
Attributes (columns)	17	96
Instances × attributes	879,189	4,637,184

Size check: $48{,}304 \times 96 = 4{,}637{,}184 \geq 3{,}000{,}000$, so the requirement is met. One-hot encoding of `location` and `cuisines` is what takes it past 3M.

Key features

Some key features, with their type (nominal, discrete or continuous) and real-world meaning. Include the response variable, if there is one.

Feature	Type	Real-world meaning
`rate`	continuous	mean customer rating out of 5; our response
`votes`	discrete	number of reviews behind the rating
`location`	nominal	Bengaluru neighbourhood, 93 levels
`cuisines`	nominal	cuisine tags, several per outlet
`cost_two`	continuous	typical bill for two, in rupees
`online_order`	nominal	takes online orders (yes/no)
`book_table`	nominal	takes table bookings (yes/no)

2. Data preparation

How did you prepare your data (encoding, outliers, missing values, duplicates), and why do you follow such steps? One row per step; the "why" matters most.

Issue found	What we did	Why, and not the alternative
`rate` is text, with `NEW` and –	parsed to a number; 3,359 `NEW` rows dropped; – set to missing	`NEW` means not yet rated, not rated zero. Coding it as 0 would invent a group of very bad restaurants.
4,399 ratings still missing	filled with the median rating of the same location, and flagged as imputed	keeps the rows for later modelling; the flag lets our tests in Section 5 leave the filled-in values out
54 duplicate chain rows	dropped on (name, address)	an outlet listed twice would inflate exactly the crowded areas we study
`cost_two` stored as "1,200"	separators removed, converted to a number	a format fix; nothing is lost
612 costs above the IQR fence	capped at the fence ($Q_3 + 1.5\,\mathrm{IQR}$)	they are real luxury outlets, so we keep the rows but stop them dominating means
`location` has 93 levels	one-hot encoded; 5 levels with under 30 rows merged into `Other`	rare levels give unstable estimates; merging keeps every crowded area intact

3. What changed after preparation

What are the changes you got after data preparation? Give before-and-after observations.

Measure	Before	After
Rows	51,717	48,304
Attributes	17	96
Missing or unusable ratings	7,758 (15.0%)	0: 3,359 dropped, 4,399 imputed and flagged
Duplicate rows	54	0
`cost_two` above the IQR fence	612	0 (capped at the fence)

Parsing `rate` turned the response from text into a number, which is what makes any test possible. Encoding `location` and `cuisines` took the attributes from 17 to 96, past the size requirement. We lost 3,413 rows: 3,359 unrated new listings and 54 duplicates.

4. Basic questions and problem statement

What are your basic questions about your data, and how is your problem statement novel from them? Show a few EDA plots, each with a one-line takeaway.

Your plot: fig_ratings.pdf

Figure 1: Ratings bunch between 3.5 and 4.2 and trail off to the left (skewness -0.7); 31% of outlets reach 4.0.

Your plot: fig_density.pdf

Figure 2: Outlets per neighbourhood run from 32 to 3,660, so a few areas would dominate a straight-line fit on density.

Basic questions

1. Do costlier restaurants get higher ratings?
2. Which cuisine has the best average rating?
3. Do outlets with more votes have higher ratings?

Our problem statements

One or two sentences each, specific to your features, and phrased so each could come out either way. Mark one as primary.

P1 (primary). Among restaurants with comparable cost and cuisine, does rating fall as the number of competing outlets in the same neighbourhood rises?

P2. Within the most crowded neighbourhoods, do outlets that take online orders rate higher than those that do not?

How they are novel: the basic questions look at each outlet on its own. Ours depend on the neighbourhood an outlet competes in, which no single row records, so they need a derived feature (outlets per neighbourhood) and a like-for-like comparison.

5. Statistical inferences

Conduct 2–3 hypothesis tests on your problem statement(s), each with parts a)–c) below. Know your tests and the reason for using them, and compare them with other tests.

Rows tested: the 43,905 outlets with an observed rating; the 4,399 imputed ratings are left out. Comparable outlets, North Indian at INR 400–800 for two: 7,877. We rank them by the number of outlets in their neighbourhood and cut them into thirds at neighbourhood boundaries: the most crowded third D (2,805 outlets in 9 neighbourhoods), a middle third (not used) and the least crowded third S (2,496 outlets in 51 neighbourhoods).

Test 1 (P1): do comparable outlets rate lower in the most crowded neighbourhoods?

a) Hypotheses, written before running the test

Before running the test, write the null and alternative hypotheses (H_0 carries the equality), the significance level, one- or two-sided, the test you will use, and the smallest difference that would matter in the real world.

H_0	$\mu_D - \mu_S \geq 0$
H_1	$\mu_D - \mu_S < 0$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Welch t-test
Smallest difference that matters	0.10 star

μ_D, μ_S: mean rating of comparable outlets in the most and least crowded thirds. One-sided, because P1 claims that ratings fall. 0.10 star, because Zomato shows ratings to one decimal: a smaller gap is invisible to a diner.

b) Other tests we considered

Which other tests did you consider, and why did you not use them? Say first why your test fits your data.

Two independent groups, a continuous response and an unknown σ call for a two-sample t-test. We use Welch's version because it does not assume equal spreads.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ; we only have sample spreads
Pooled t-test	assumes equal variances, which an F-test rejects ($F = 1.31$, $p < 0.001$). Run anyway, it gives $t = -4.96$ and the same decision.
Paired t-test	nothing to pair: each outlet is in one group
t-test on the correlation r	sees only a straight-line trend, which Figure 2 shows a few large areas would drive

c) Result

Report the test statistic, degrees of freedom, p-value and decision, and the estimate with a 95% confidence interval. Is the difference large enough to matter?

Estimate	-0.06 star (3.68 in D, 3.74 in S)
95% confidence interval	-0.084 to -0.036
Test statistic (d.f.)	$t = -4.92$ (4,984)
p-value	4.4×10^{-7}
Decision	reject H_0

Comparable outlets in the most crowded neighbourhoods rate 0.06 star lower on average. The difference is real, but it does not matter by our standard: the whole interval lies inside ± 0.10 star.

Test 2 (P2): does online ordering lift ratings in the most crowded neighbourhoods?

The same three parts as Test 1.

a) Hypotheses, written before running the test

H_0	$\mu_{\text{on}} - \mu_{\text{off}} \leq 0$
H_1	$\mu_{\text{on}} - \mu_{\text{off}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Welch t-test
Smallest difference that matters	0.10 star

μ_{on}, μ_{off}: mean rating, within D, of outlets that do and do not take online orders. One-sided, because P2 claims a lift.

b) Other tests we considered

Again two independent groups, a continuous response and an unknown σ, so a two-sample t-test; we fixed Welch's version in advance.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ; we only have sample spreads
Pooled t-test	also valid here, since an F-test does not reject equal variances ($F = 1.10$, $p = 0.08$); it gives $t = 1.25$ and the same decision. We keep Welch, fixed in advance.
Paired t-test	no outlet is in both groups

c) Result

Estimate	$+0.02$ star (3.69 with online orders, 3.67 without)
95% confidence interval	-0.012 to $+0.052$
Test statistic (d.f.)	$t = 1.24$ (2,086)
p-value	0.11
Decision	fail to reject H_0

Our data gives no evidence that taking online orders lifts ratings in crowded neighbourhoods. The interval rules out a lift larger than 0.052 star, so any lift is below the 0.10 star that would matter.

Running a third test? Add it here as Test 3, with the same three parts: copy the Test 2 block.

6. Limitations

What your analysis cannot show, and why: rows you dropped, assumptions that hold only approximately, groups your conclusions do not cover, and association versus cause.

- **Association, not cause.** Crowded neighbourhoods also differ in rent, footfall and customers; our tests cannot separate density from these.
- **Outlets are not fully independent.** Outlets in one neighbourhood share conditions, so our effective sample is smaller than 5,301 and our p-values are optimistic. A test on the 60 neighbourhood averages would be stricter.
- **Established outlets only.** The 3,359 unrated new listings were dropped in Section 2.
- **Imputed ratings left out.** The 4,399 outlets with a missing rating may differ from the rest, and they are not in our tests.
- **One cuisine and price band.** We compared North Indian outlets at INR 400–800 for two; other cuisines and prices may behave differently.